

RCC-OMP FINAL IMPLEMENTATION ROADMAP

Locked requirements • module changes • backend architecture • exact PDF/SRB strategy • implementation order • estimates • testing

Current baseline: System Inventory is the asset model. Machines, Department and a separate Reports module are not required. Notification Trash work, Notes changes, recurring lifecycle fixes, User Management implementation and Help Center documentation have already been completed in the current prototype. This roadmap covers the new post-demo requirements.

Important: This document does not assume an unavailable backend or an unavailable SRB template. Where the exact production/deployment input is missing, it is explicitly marked as an input.

1. LOCKED FINAL REQUIREMENTS

Positions

- Exactly two positions: **Head** and **IT Executive**.
- Heads can do everything in the application.
- IT Executives can do everything operationally.
- Any eligible user may verify another user's task.
- **Self-verification is never allowed:** completedBy must differ from verifiedBy.
- When one Executive completes a task, eligible other users receive a verification notification.

User Management

- Create accounts directly from User Management.
- Choose Head or IT Executive while creating the account.
- Add, deactivate and reactivate users.
- Reset/change another user's password from the user row.
- Generate a random temporary password and download it as a .txt file containing Employee ID + temporary password.
- First login after reset forces a password change.
- Password expires every 60 days.
- Normal application login uses Employee ID + password; remove normal user email identity/login.
- Retain one common emergency account only for the exceptional situation where all accounts require password recovery/reset.

Login security

- Three failed login attempts → account lock.
- Successful login resets failed-attempt count.
- Password expiry and reset are enforced by the authentication/service layer in production.
- Do not store plaintext passwords.

2. COMMON TASK + VERIFICATION WORKFLOW

Use one shared workflow instead of separately inventing completion/verification rules in each module.

CREATE/SCHEDULE → ACTIVE → COMPLETED BY A → VERIFICATION PENDING → VERIFIED BY B → PDF DOWNLOAD (where applicable)

- Tasks may be completed **before** their due date. The due date is a deadline, not the only completion day.
- Completion stores employee ID and timestamp.
- Completion creates a verification-pending state and notification.
- The completer cannot verify the same record.
- Verification stores verifier ID and timestamp.
- PDF becomes downloadable after verification where that module requires it.
- Undo Completion returns the task to the correct active state while preserving the audit trail.
- Undo, completion and verification all create History events.

Central business rule

completedBy == verifiedBy → reject verification

This rule must be enforced in the backend/service layer, not only by hiding a button in React.

3. MODULE CHANGE MATRIX

Module	Final work	Priority
User Management	Two positions; account creation; reset/change password; deactivate/reactivate; Employee ID login support; 60-day expiry; 3-attempt lock; temporary .txt password; remove normal email.	P0
Backup	Add No. of files field; completion → verification; notification; verified PDF download; undo; History.	P0
QA	Verification; early completion; undo; reminder snooze/loop; verification notification; History.	P0
Preventive Maintenance	Verification; early completion; undo; reminders/snooze; History.	P0
Inspection	Verification; early completion; undo; reminders/snooze; History.	P0
Monthly Hard Disk	Dispatch only 1–7; Return only 28–31; verification; early completion where applicable; undo; reminders; History.	P0
Notifications	Keep Trash/deleted lifecycle; shared snooze/reschedule; verification alerts; preserve task evidence.	P0
Notes	Keep pin/unpin, pinned creation, search/folder/view/edit/save/delete and user-scoped persistence; Share Link remains removed. Manual regression test required.	P1
System Inventory	Remain the asset model. Do not reintroduce Machines.	P1
History	NEW central module visible to all users; name + Employee ID + module + action + date/time + record/details.	P0
Help Center	Already updated to current app; extend only after final rules/templates are locked.	P2
Settings	Keep implemented notification controls; no normal email-notification/critical-only controls.	P2

4. REMOVED — DO NOT REINTRODUCE

- Machines module — removed; System Inventory replaces it.
- Department module — not required.
- Separate Reports module — not required; PDFs/reports belong inside the relevant module.
- Old three-role model — replaced by Head + IT Executive.
- Extra IT role variants — not required.
- Separate Permission Matrix screen — not required.
- Separate Access Log module — History replaces it.
- Invite-sent flow — not required.
- Normal user email IDs/login — removed.
- Notes Share Link — remains removed.
- Invented Notes source/module categories — do not create.
- Completion-only-on-due-date restriction — removed.

5. REMINDERS + SNOOZE

Implement this as one shared reminder service used by QA, Backup, PM, Inspection and Hard Disk.

- Support a chosen future reminder date.
- Support recurring reminder intervals such as every 10 days where configured.
- Snooze/reschedule changes the next reminder without changing the task's actual due date.
- Completion/verification/undo must update reminder state correctly.
- Verification-pending notifications go to eligible other users.
- Deleted notifications use the existing Trash lifecycle rather than destroying task evidence.

Example: QA deadline in one month → reminder every 10 days → snooze/reschedule if needed → complete → verification notification → another user verifies → PDF becomes available.

6. HARD DISK DATE RULES

Action	Allowed	Invalid
Dispatch	1st–7th of month	8th onward
Return	28th–31st of month	1st–27th

Validate both in UI and in the service/business layer.

7. HISTORY / AUDIT MODULE

History is a new central module and is visible to all users.

- Login success/failure/lockout/logout where available.
- User created, deactivated, reactivated and password reset.
- Task created/scheduled/edited.
- Task completed.
- Verification requested and verified.
- Completion undone.
- Reminder created/snoozed/rescheduled/cleared.
- Important notification lifecycle actions.

- PDF generated/downloaded.
- Important settings/configuration changes.

Minimum event record

eventId timestamp employeeId userName module action recordId details
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8. BACKEND ARCHITECTURE

The new security requirements should not remain frontend-only. LocalStorage can be retained only where appropriate for prototype-only UI state; authentication, password expiry, lockout, sessions, verification authority and audit history need a trusted service/database in production.

Recommended production flow

React/Vite → Backend API/service layer → PostgreSQL database + private object storage → Auth/session + notification/reminder + audit + document services

Core entities

Entity	Important fields
users	id, employeeId, name, position, status, passwordHash, passwordChangedAt, passwordExpiresAt, failedAttempts, lockedUntil, mustChangePassword
sessions	sessionId, userId, loginAt, lastSeenAt, logoutAt, active, device/browser metadata
task records	module-specific fields + createdBy + completedBy + verifiedBy + timestamps + status + dueAt
notifications	recipientId, type, taskId, createdAt, readAt, snoozedUntil, deletedAt, status
audit_history	eventId, actorId, module, action, recordId, timestamp, details
document_templates	templateId, module, version, storagePath, checksum, active, effective date
generated_documents	documentId, recordId, templateId/version, generatedBy, generatedAt, storagePath

Do not hard-code authorization into individual buttons. The service layer should determine whether the current employee can complete/verify/reset/manage.

9. PASSWORD + SESSION FLOW

Normal login

Employee ID + password → authenticate → check lock → check password expiry → create session → application access.

Reset flow

- Authorised manager selects user in User Management.
- Backend generates cryptographically random temporary password.
- Backend stores only a secure hash.
- System downloads .txt containing Employee ID + temporary password.
- User logs in with Employee ID + temporary password.
- mustChangePassword forces password change before normal access.
- New password resets the 60-day clock.
- Old sessions/credentials for that account should be revoked according to policy.

Emergency common account

The confirmed requirement is one common account for the exceptional case where all accounts require password recovery/reset and normal access is unavailable. The exact credential custody, MFA and recovery procedure is a deployment/security decision and should not be guessed in code.

10. ACTIVE SESSION / PC VISIBILITY

The website can record active sessions, Employee ID, name, login time, last-seen time and available browser/device metadata.

A normal browser cannot reliably expose the Windows computer hostname. If you specifically need a name such as a Windows PC hostname, use a controlled enterprise mechanism (for example a trusted desktop agent or managed-device integration). Do not fabricate a PC name from browser information.

11. EXACT SRB / PDF SOLUTION

Do not store the actual SRB PDF binary inside PostgreSQL. Store the original template in private object/file storage and store its metadata/version/path in the database.

Exact-format workflow

- Obtain the official original SRB template. Prefer the original PDF supplied by the organisation.
- Upload that original PDF to private object storage.
- Store template ID, module, version, storage path, checksum, active/effective dates in the database.
- If the PDF has AcroForm fields, fill those fields.
- If it is a fixed non-fillable PDF, use the original PDF as the background and overlay dynamic text at mapped coordinates.
- Do not redraw the official layout in HTML/CSS when exact visual matching is required.
- Version templates instead of overwriting them.
- Store generated PDFs with record ID + template version + generation timestamp.

Database vs file storage

Database	Private file/object storage
Template ID, module, version	Original SRB PDF
Storage path/reference	Generated PDFs
Checksum/hash, active flag, dates	DOCX/image assets if required

Important: the handwritten/photo notes define requirements and fields, but they cannot guarantee exact PDF layout. For exact formatting, you must supply the official SRB PDF/template file. Once supplied, its fields/coordinates can be mapped exactly.

12. IMPLEMENTATION ORDER + ESTIMATE

Phase	Work	Estimate*
0	Repo audit + freeze final requirements + map existing auth/storage	0.5–1 day
1	Backend foundation: DB, API/service boundaries, users, sessions	1–2 days
2	User Management + password expiry/reset/lockout	1–1.5 days
3	Shared completion/verification + permissions + verification notifications	1–2 days
4	Backup + QA + PM + Inspection + Hard Disk integration	2–3 days
5	Shared reminder snooze/loop	1–1.5 days
6	History + audit integration	1–1.5 days
7	SRB template/PDF infrastructure + module PDFs	1–3 days after templates
8	Sessions/device visibility + hardening	0.5–1.5 days
9	Manual testing + regression + fixes	1–2 days

*Planning estimates only. Exact duration depends on the current repo, whether a real backend already exists, deployment environment and the actual SRB templates.

13. EXACT IMPLEMENTATION PROMPTS

Use these sequentially. After each implementation prompt: run TypeScript check, production build and git diff --check; inspect the diff; then commit.

Prompt 1 — Backend foundation

Audit the current RCC-OMP repository and implement the backend-ready foundation for Users, Sessions, module task records, Notifications, Audit History and Document Templates. Do not reintroduce Machines, Departments or a Reports module. Do not invent a backend provider if one is not configured. Report what is truly available before implementing.

Prompt 2 — Final User Management

Update User Management to exactly two positions: Head and IT Executive. Allow account creation directly in User Management, position selection, deactivate/reactivate, search/filter and reset/change password actions. Remove normal user email identity/login. Do not add extra roles or a Department module.

Prompt 3 — Password security

Implement Employee ID login, 60-day password expiry, 3-failed-attempt lockout, secure random temporary password, forced first-login password change and admin reset. Download a .txt containing Employee ID + temporary password. Never persist plaintext passwords. If the repo has no real backend, clearly separate prototype behavior from server-enforced security.

Prompt 4 — Shared verification engine

Create one reusable completion/verification workflow. Any eligible user can complete; any other eligible user can verify; self-verification must be rejected in the service layer. Record completedBy/completedAt/verifiedBy/verifiedAt. Allow completion before due date. Add Undo Completion while preserving audit history.

Prompt 5 — Verification notifications

Integrate verification-pending notifications into the existing notification system. When a user completes a task, notify eligible other users. Preserve the existing Trash/deleted-notification lifecycle and never delete underlying task evidence.

Prompt 6 — Backup

Add required 'No. of files' to Backup. Apply completion → verification → verified PDF download, prevent self-verification, add Undo Completion and History while preserving existing recurrence/reminder behavior.

Prompt 7 — QA/PM/Inspection/Hard Disk

Apply the shared completion, verification, early-completion, undo and audit workflow to QA, Preventive Maintenance, Inspection and Monthly Hard Disk Tracker without duplicating date logic.

Prompt 8 — Hard Disk rules

Enforce Monthly Hard Disk Tracker Dispatch only on days 1–7 and Return only on days 28–31. Validate in UI and service/business layer. Do not add Machines.

Prompt 9 — Reminder snooze/loop

Implement one shared reminder snooze/reschedule mechanism for QA, Backup, PM, Inspection and Hard Disk. Support a chosen future reminder date and recurring intervals such as every 10 days where configured. Update reminder state correctly after completion, verification and undo.

Prompt 10 — History

Build a system-wide History module visible to all users. Record employee ID, name, module, action, record ID, timestamp and relevant details for user management, login/security, task completion, verification, undo, reminders and PDF generation/download.

Prompt 11 — SRB/PDF

Implement template metadata/versioning and PDF generation without inventing an SRB layout. Use the supplied official SRB PDF as background or its AcroForm fields. If the actual template is missing, stop before claiming exact formatting and report exactly which template file is required.

Prompt 12 — Active sessions

Implement active-session visibility showing name, Employee ID, active status, login time, last-seen time and available device/browser metadata. Do not fabricate a Windows PC hostname; document any trusted-agent/managed-device dependency.

Prompt 13 — Final audit

Audit the complete repo against the locked requirements. Confirm Machines, Department and separate Reports are absent/not reintroduced; normal user emails are removed; roles are exactly Head and IT Executive; verification, undo, reminders and History are consistent.

Prompt 14 — Final manual test prep

Do not add features. Produce the complete manual browser testing checklist for login security, User Management, every module, completion, verification, notifications, snooze, undo, History, PDF generation and sessions. Run TypeScript, build and git diff --check.

14. FINAL MANUAL TESTING CHECKLIST

Area	Test
Login	Correct/wrong Employee ID; wrong password; attempts 1/2/3; lock; unlock/recovery; expiry; forced reset; logout.
Users	Create Head; create Executive; duplicate ID; deactivate/reactivate; reset; .txt; first-login change.
Verification	A completes → B verifies; A cannot verify; Head/Executive cross-verification; notification; timestamps; PDF.
Undo	Complete → undo → active; no duplicate recurrence; reminder state correct; History contains both events.
Backup	No. of files; completion; verification; PDF; undo; refresh persistence.
QA/PM/Inspection	Early completion; due boundary; verification; reminder; snooze; undo; recurrence; refresh.
Hard Disk	Dispatch 1–7 valid, 8+ invalid; Return 28–31 valid, other dates invalid.
Notifications	Reminder; verification alert; snooze; recurring reminder; Trash; task evidence remains.
Notes	Create/edit/delete/search/folder/view; pin/unpin; create in Pinned; reload; user isolation; Share Link absent.
History	Correct name/Employee ID/module/action/time; all users can see complete history.
PDF	Exact template; correct fields; dates/times; verifier; download; template version.
Sessions	Active session appears; logout ends session; last-seen updates; device data is truthful.

FINAL RECOMMENDATION

Do not continue with isolated frontend patches for these requirements. Build the shared authentication, task-verification, reminder, audit-history and document foundations first; then connect each module to them. This will prevent five different implementations of the same rules.